

Enterprise Computing HSC 2025 – sample full mark responses

Question 14 (3 marks)

A company wants to analyse memes on social media to assess the effectiveness of its advertising campaign.

Describe how memes can be analysed to provide insights into the effectiveness of the advertising campaign.

Sample

memes provide insights into advertising effectiveness by looking at reach and audience they attract and whether that works for the business. Analysis includes content type, humor, popularity, visuals, engagement/views. and also whether the business who is advertising recieved an increase in sales, interest or followers.

Question 16 (8 marks)

A TV manufacturer wants to improve the efficiency and reliability of its manufacturing system. It plans to incorporate an intelligent system into its manufacturing process to collect live data for analysis. Hardware devices will be installed to collect data on the system's operations. The data is then sent to a server for processing.

The manufacturing processes include attaching the LCD panel to the motherboard and testing the brightness of the screen. The current system is controlled through manual processes and monitoring.

- (a) Describe the hardware that is needed for the intelligent system. Include examples of specific types of sensors in your answer. (4 marks)**

Sample

The hardware that is needed for the intelligent system include a light sensor and a voltage sensor. A light sensor is needed as it will measure the brightness of the screen and ensure it passes a certain standard, whilst also sending the individual data recordings of how bright each screen to a server for processing, finding trends like the rate of screens that aren't bright enough, the average brightness of each screen, etc. Additionally, a voltage sensor will ensure that electricity can flow through the TV and it can function properly, as if there is an issue with wiring or anything internal, then the sensor will pick up that the flow of electricity has halted. This will ensure less defects going out to shop front, and can catch them early in the manufacturing process to ensure that they don't cause further issues down the line. The data from this sensor will be sent to a server for processing finding trends such as defect rates.

- b) Explain the potential benefits of introducing an expert system into the manufacturing process. Support your answer with reference to Industry 4.0. (4 marks)**

Sample

Industry 4.0 refers to the automation of general manufacturing and simple repetitive tasks through the use of expert systems and AI. The potential benefits of introducing this system into the manufacturing process are decreased defect rates, increasing efficiency, and lower costs. Expert systems and AI are less likely to generate mistakes, as there is no room for human error, meaning the company will have a lower rate of defect TV's which will allow for less expenses and greater profits. Additionally, expert systems can function at a much faster and more efficient rate than humans can due to constant improvements and updates, meaning greater manufacturing capabilities in a shorter time. Lastly, using one large scale expert system instead of a multitude of labour resources can be significantly cheaper in the long run, as the upfront cost of the system will be cheaper than the outgoing cost of continually paying labour, which save costs for the business.

Question 17 (8 marks)

A coffee shop chain has branches in all major cities in Australia. It collects a large volume of sales data as well as geographic and demographic data from each of its coffee shop branches.

The chain plans to use a cloud-based data warehouse for storage and for future analysis.

(a) Explain how the use of data warehousing could benefit this coffee shop chain. (3 marks)

Sample

Data warehousing is where large data can be stored, analysed and processed. The coffee shop chain can benefit off this by using the processed data from the data warehouse to analyse their customers behaviours in each coffee shop branches, allowing them to identify patterns and trends, as well as making future analysis and predictions. Using a cloud-based data warehouse can also be beneficial for all of the coffee shop branches as they can store all their data in a single remote cloud storage server, which can be more efficient for analysing and processing data.

(b) Explain how the advancement of hardware can affect the processing of data from the data warehouse for this coffee shop chain. (3 marks)

Sample

Advancements in hardware such as processing power can allow for the organisation to analyse data from the datawarehouse more quickly, increasing the volume and velocity at which it is processed. This would allow the organisation to discover trends and patterns more efficiently, leading to better decision making. Advancements in storage and memory can also allow for the data warehouse to store more data at a time, increasing its volume and variety of stored data, such as comparing data from multiple branches or timeframes at once. This in turn can be analysed to provide deeper insights into data, increasing the quickly at which data is processed, and allowing the coffee chain to make even better decisions and predictions. Finally, advancements in communication media involves improvements to cloud storage media, which the coffee shop chain is planning to use. This advancement would further enhance the above for a cloud storage data warehouse in securely storing and analysing data in an accessible cloud location.

Question 18 (4 marks)

A small car rental business, with five vehicles, wants to use a spreadsheet to keep track of its bookings.

A staff member will enter the driver's name, vehicle size selected (small, medium, large), and the number of days of rental for each vehicle.

The spreadsheet should automatically provide the following information:

- daily rate – small (\$200), medium (\$250), large (\$300)
- discount – a discount of 20% is allowed if the booking is for at least 7 days
- rental cost – calculated by multiplying the daily rate and the number of days, and allowing for a discount if applicable.

Part of the spreadsheet has been set up. Complete the spreadsheet so that the required information will automatically be displayed when rentals are entered.

Sample 1

Student input

	A	B	C	D	E	F	G
1	Vehicle bookings						
2	Vehicle	Driver	Vehicle size	Daily rate (\$)	No. of days	Discount (%)	Rental cost (\$)
3	1	XX	Small	=VLOOKUP(C3,A\$10:B\$12,2)	7	=IF(E3>=7,"20","0")	=IF(F3=20, D3*E3*0.8, D3*E3)
4	2	XX	Large	=VLOOKUP(C4,A\$10:B\$12,2)	9	=IF(E4>=7,"20","0")	=IF(F4=20, D4*E4*0.8, D4*E4)
5	3	XX	Medium	=VLOOKUP(C5,A\$10:B\$12,2)	5	=IF(E5>=7,"20","0")	=IF(F5=20, D5*E5*0.8, D5*E5)
6	4	XX	Large	=VLOOKUP(C6,A\$10:B\$12,2)	7	=IF(E6>=7,"20","0")	=IF(F6=20, D6*E6*0.8, D6*E6)
7	5	XX	Small	=VLOOKUP(C7,A\$10:B\$12,2)	4	=IF(E7>=7,"20","0")	=IF(F7=20, D7*E7*0.8, D7*E7)
8							
9	Vehicle size	Daily rate (\$)					
10	Small	200					
11	Medium	250					
12	Large	300					

Student output

	A	B	C	D	E	F	G
1	Vehicle bookings						
2	Vehicle	Driver	Vehicle size	Daily rate (\$)	No. of days	Discount (%)	Rental cost (\$)
3	1	XX	Small	200	7	20	1120
4	2	XX	Large	300	9	20	2160
5	3	XX	Medium	250	5	0	1250
6	4	XX	Large	300	7	20	1680
7	5	XX	Small	200	4	0	800
8							
9	Vehicle size	Daily rate (\$)					
10	Small	200					
11	Medium	250					
12	Large	300					

Sample 2

Student input

	A	B	C	D	E	F	G
1	Vehicle bookings						
2	Vehicle	Driver	Vehicle size	Daily rate	No. of days	Discount	Rental cost
3	1		small	=IF(C3="small", "200", IF(C3="medium", "250", IF(C3="large", "300")))	1	=IF(E3>="7", "Yes", "No")	=IF(F3="Yes", D3*E3*0.8, D3*E3)
4	2		medium	=IF(C4="small", "200", IF(C4="medium", "250", IF(C4="large", "300")))	8	=IF(E4>="7", "Yes", "No")	=IF(F4="Yes", D4*E4*0.8, D4*E4)
5	3		large	=IF(C5="small", "200", IF(C5="medium", "250", IF(C5="large", "300")))	7	=IF(E5>="7", "Yes", "No")	=IF(F5="Yes", D5*E5*0.8, D5*E5)
6	4		small	=IF(C6="small", "200", IF(C6="medium", "250", IF(C6="large", "300")))	15	=IF(E6>="7", "Yes", "No")	=IF(F6="Yes", D6*E6*0.8, D6*E6)
7	5		large	=IF(C7="small", "200", IF(C7="medium", "250", IF(C7="large", "300")))	20	=IF(E7>="7", "Yes", "No")	=IF(F7="Yes", D7*E7*0.8, D7*E7)

Student output

	A	B	C	D	E	F	G
1	Vehicle bookings						
2	Vehicle	Driver	Vehicle size	Daily rate	No. of days	Discount	Rental cost
3	1		small	200	1	No	200
4	2		medium	250	8	Yes	1600
5	3		large	300	7	Yes	1680
6	4		small	200	15	Yes	2400
7	5		large	300	20	Yes	4800

Question 19 (3 marks)

An education business intends to provide a subscription service to teachers and students for online resources. To register, the subscriber needs to provide their name, email address and date of birth, and indicate whether they are a teacher or student.

Design a user interface for registering an account. Clearly label all features.

Sample

menu buttons leading to other pages

logo Home About Resources search Login

Email address

Name

Password

Confirm password

Date of birth

Which are you?

Register

buttons to hide password

drop down box to select whether user is a student or teacher

Question 21 (5 marks)

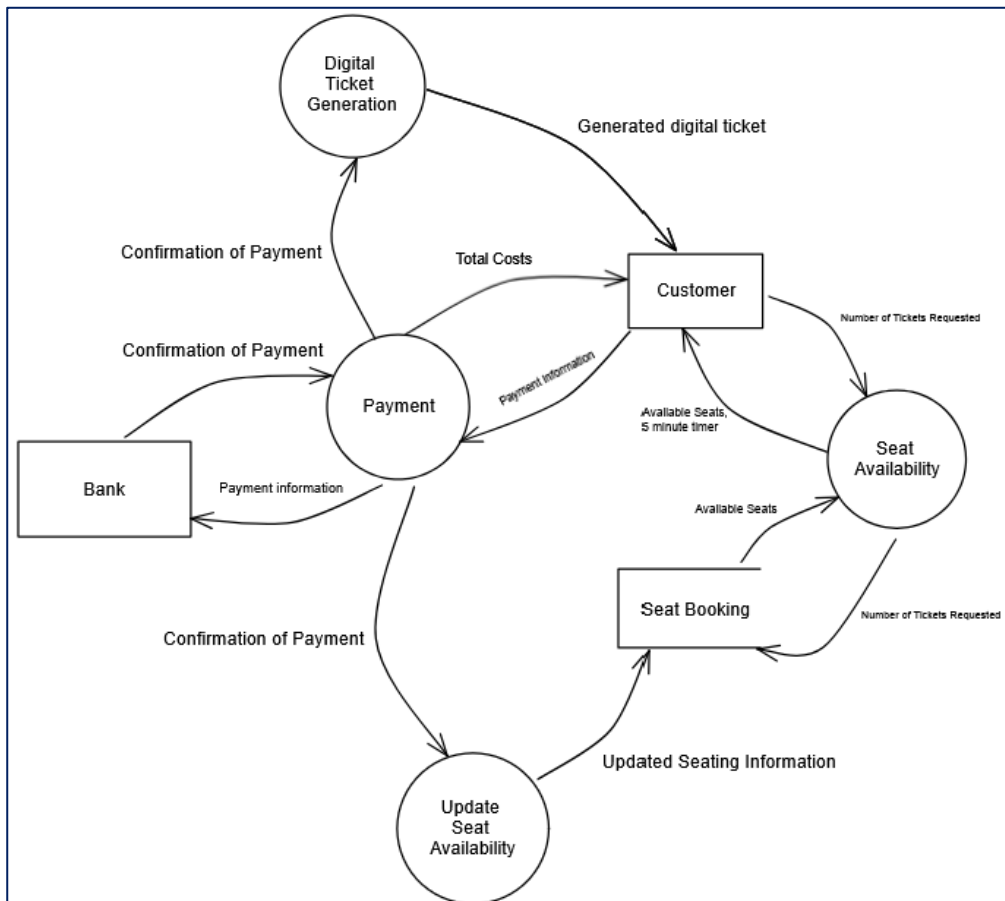
An online booking system allows customers to purchase tickets to a concert. Once a customer specifies the number of tickets required, the system displays the seats available and allows the customer to make seat selections.

The selected seats are held for five minutes for the customer to proceed to payment. The system displays the total cost and allows the customer to pay using their credit card.

After payment is confirmed by the bank, a digital ticket is generated and sent to the customer. The seat booking database is also updated.

Construct a data flow diagram (DFD) to represent this system.

Sample



Question 22 (5 marks)

An insurance company hires a project team to develop a system that manages the personal information of customers and employees, customer claims and business contracts. The data will be stored and accessed on a cloud-based server and backed up locally.

(b) Justify TWO methods that the company can use to maintain the security of the data. (3 marks)

Sample

The company can launch a training program for the employees. This program can do two things - train the employees about the importance of customer and data privacy, encouraging them to keep the information secure. It can also act as a proactive measure against data breaches, such as phishing scams, to prevent any leaks and breaches for the customer data. The company can also utilise two factor authentication for the people who will be able to access the data. This will ensure that whoever is trying to access the data is someone with authorization, and provide a digital record of who accessed it to persuade against any security breaches.

Question 23 (7 marks)

As a result of advancements in technology, a company now presents employee performance information on a dashboard that is updated using real-time data, rather than spreadsheets. The data dashboard is shown.

(The data dashboard can be viewed on the [2025 Enterprise Computing HSC online exam](#).)

(a) Describe how advancements in software have empowered data analytics. In your answer, refer to the scenario. (3 marks)

Sample

Advancements in software have empowered data analytics as by updating the system under industry 4.0, the business is able to automate processes and allow for real time feedback and analytics which, in turn provide more in depth information on the performance on the business, allowing them to focus on their efficiency in other key areas. Instead of using spreadsheets, and manually uploading data to track performance, wasting unnecessary amounts of time, they use a constantly updated data dashboard that visually highlights the key information such as employee performance in each department, signifying where the business needs to implement more resources, allowing for a more efficient and well performing business.

(b) Explain how the visualisations used in the data dashboard provided could affect the management team's user experience and understanding of employee productivity. (4 marks)

Sample

The visualisations in the data dashboard are easy to read, catering to users of all digital literacy levels. This will positively affect the management team as not all members of the management team will be well versed in understanding data analytics. Hence, by presenting information from the data with visuals that are familiar, simple and have a clean aesthetic and presentation, the entirety of the management team will be able to read and understand what trends and information has been found in the data through the visuals. Because the management team will have a positive user experience when accessing the dashboard through the visualisations, the team will be able to better understand the trends within their employees productivity. Through colour-coding and simple visuals the team can quickly and easily determine which department has the highest/lowest performance, which months reap the highest/lowest productivity, which department during which months have the highest/lowest employee attendance and which employees are currently performing best. Because the visualisations are dynamic due to the real-time updates of the dashboard, the management team can also track in real-time any changes that occur to employee productivity and then identify what could have caused the change. This will allow the management to understand what factors, such as season, month or attendance patterns, will boost or decrease employee productivity and how they can better manage these factors to achieve higher productivity rates and performance.

Question 24 (5 marks)

A food company is considering the use of freelance work and offshore development to support its packaging design.

Explain the advantages and disadvantages to the company in using freelance work and offshore development for packaging design.

Sample

There are various factors to consider when hiring freelance work for the development of packaging design. The advantage of hiring a freelance designer is that the company obtains access to specialised skills. By resorting to an external source for package designing, they are saving the expenses on the costs involved in training one of their own employees to conduct the task themselves. This allows the company to invest their funds into other operational processes, optimising their allocation of resources. On the other hand, however, the freelance designer may require negotiation in terms of usage rights of their intellectual property, as a contract will be required to be signed upon agreement to collaboration, which may result in a loss of flexibility for the company in the designs they are provided.

There are also a multitude of considerations associated with offshore development. An advantage that the company will have by employing offshore development as a part of their business operations is that they are given access to a global talent pool which may extend over the scope of domestic talent. Despite this, time-zones can become a disadvantage when meetings are required, as communication between international sites may become delayed, potentially slowing down operations.

Question 25 (8 marks)

View the slideshow about applications of decision support systems.

(The slideshow can be viewed on the [2025 Enterprise Computing HSC online exam](#).)

Explain how intelligent systems can assist the different categories of decision-making. Support your answer with some examples from the stimulus.

Sample

Intelligent systems can support the 3 main types of decision-making.

Structured decisions are decisions that **can be fully automated by computers**. These decisions can be covered by a set of 'if' statements and do not require much context or human intuition, and there is limited risk of the system messing up. Here, expert systems (such as a DSS [decision support system]) can be used to gather input data and use algorithms and known knowledge to convert it into an automated process. For example, **fraud prevention** in finance can be structured by monitoring user accounts and automatically blocking transactions if they seem fraudulent. Another example is **scheduling and planning** in healthcare, where systems can use existing schedules as well as information from patient visits to determine the most optimal rostering for efficiency.

Semi-structured decisions are decisions that **can be partially automated but still need some human intervention**. These decisions have certain aspects that can be covered by 'if' statements in an intelligent system (such as a DSS) but still need human intervention to prevent any mishaps or provide background context. They are also usually more high-risk where human intuition is needed. An example of this is **demand forecasting** in transportation, where the system can analyse past data to predict future demand, but humans are still required to use the system's decisions to actually make real change. Another example is **loan applications** in finance, where the system can use data such as credit score to provide additional support to a human, but cannot make the decision on their own due to risk of bias or fraud.

Finally, unstructured decisions are decisions that **cannot be automated by computers** and require humans to work. These decisions require lots of context or heuristic (experience-based) knowledge and are usually high-risk where human intuition is definitely required. However, these decisions can still be supported using systems such as a DSS. An example of this is **medical diagnosis**, where the knowledge and experience of a doctor is required in order to actually make proper diagnosis, but systems that can analyse scans for example can assist the doctor in making their judgement. Another example is **HR recruitment**, where a human is still required for interviews and final decision making, but systems can help check resumes, perform background checks, and provide suggestions on what decision to make.

In conclusion, intelligent systems can support all 3 types of decisions. For structured decisions, they can fully automate the task, reducing the need for employees. For semi-structured decisions, they can assist with automation but still require some human intervention, but they significantly reduce the load on employees. For unstructured decisions, they can't automate the decision making process but they can assist the employee with making their decision.